

**Exp no :08**

**Date :**

## **Use Steg-Expose to Detect Hidden Data in Images**

### **Description**

**StegExpose** is a specialized tool used for **steganography analysis**. It works by evaluating the **statistical properties** of an image to estimate the probability of hidden data being embedded within it. This process helps forensic investigators detect steganography techniques like Least Significant Bit (LSB) embedding.

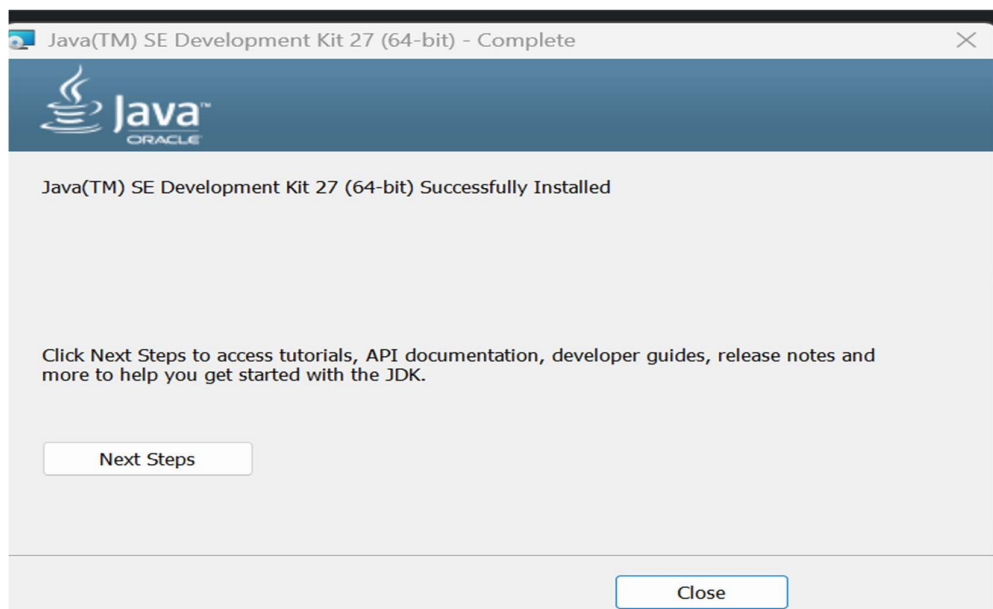
### **Prerequisites**

- Java Runtime Environment (JRE): StegExpose is a Java-based application and requires JRE to run.
- StegExpose Tool: Download the latest .jar file from the official StegExpose GitHub repository.

### **Step-by-Step Process**

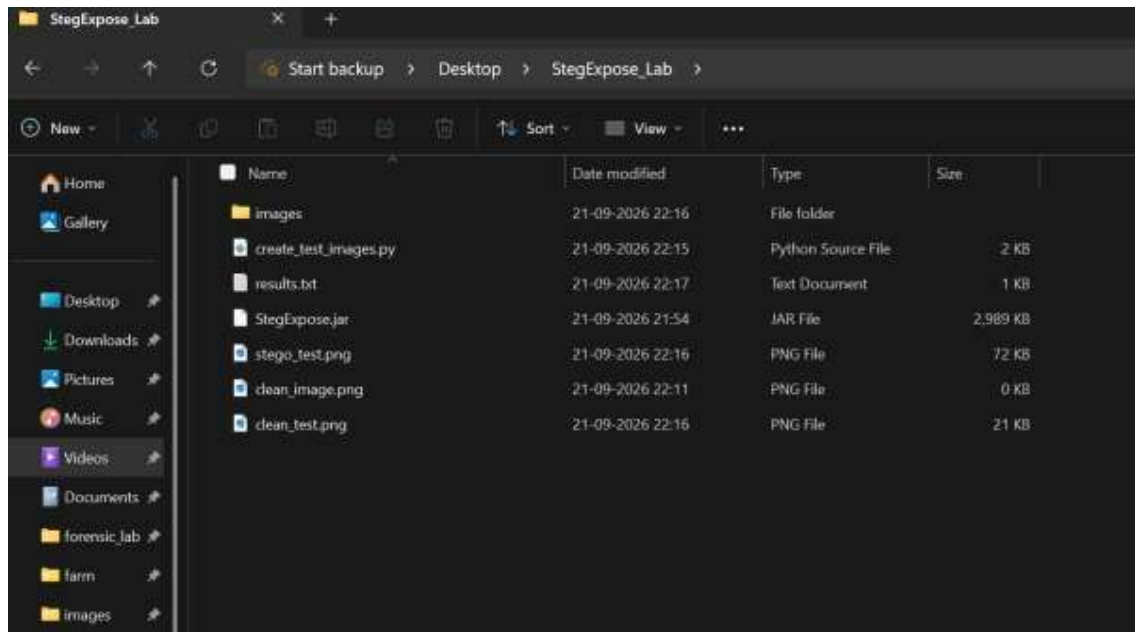
#### **1. Download and Set Up StegExpose**

1. Download the tool: Obtain the StegExpose.jar file from the GitHub repository.
2. Install Java: Ensure JRE is installed on your machine.
3. Prepare environment: Place the StegExpose.jar file in a dedicated working folder.



## 2. Select Images for Analysis

- Collect the images you suspect might contain hidden data.
- StegExpose supports common image formats such as .png, .jpg, and .bmp.



## 3. Open Command Line or Terminal

- Navigate to the folder where the StegExpose.jar file is located using your Command Prompt (Windows) or Terminal (Linux/macOS).

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\stegexpose> dir

Directory: C:\stegexpose

Mode                LastWriteTime         Length Name
----                -
d-----          22-09-2026   10:28             images
-a----          22-09-2026   10:27       3060550 StegExpose.jar

PS C:\stegexpose> |
```

#### 4. Run StegExpose on an Image

- Use the following command structure to analyze a single image for hidden data:

#### Command Structure:

java -jar StegExpose.jar <image\_file\_path> **Example:**

java -jar StegExpose.jar test\_image.png

#### 5. Analyze the Output

- StegExpose calculates a "**suspect**" score ranging from 0 to 1.  
**The higher the score, the more likely steganography is present.**

| Score Range   | Interpretation (Suggested Thresholds)                       |
|---------------|-------------------------------------------------------------|
| Less than 0.2 | Image is considered <b>clean</b> (no hidden data detected). |
| 0.2 - 0.3     | <b>Possibly</b> some hidden data is present.                |
| Above 0.3     | <b>Likely</b> that steganography is present.                |

- **Example Output Analysis:** java -jar StegExpose.jar

suspect\_image.png

Analyzing suspect\_image.png...

Result: 0.4

Steganography likely present

```
PS C:\stegexpose> java -jar StegExpose.jar images
PS C:\stegexpose> java -jar StegExpose.jar images default 0.2 results.csv
PS C:\stegexpose> dir
```

Directory: C:\stegexpose

| Mode    | LastWriteTime    | Length  | Name           |
|---------|------------------|---------|----------------|
| d-----  | 22-09-2026 10:28 |         | images         |
| -a----- | 22-09-2026 10:36 | 270     | results.csv    |
| -a----- | 22-09-2026 10:27 | 3060550 | StegExpose.jar |

```
PS C:\stegexpose> |
```

```

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PS C:\Users\bharg> cd C:\stegexpose
PS C:\stegexpose> dir images

    Directory: C:\stegexpose\images

Mode                LastWriteTime         Length Name
----                -
-a-----         08-09-2026         09:51       213535 Screenshot 2026-09-08 095126.png

PS C:\stegexpose> java -jar StegExpose.jar images
PS C:\stegexpose> java -jar StegExpose.jar images default 0.2 results.csv
PS C:\stegexpose> dir

    Directory: C:\stegexpose

Mode                LastWriteTime         Length Name
----                -
d-----         22-09-2026         10:28       images
-a-----         22-09-2026         10:36         270 results.csv
-a-----         22-09-2026         10:27     3060550 StegExpose.jar

PS C:\stegexpose> notepad results.csv

```

## 6. Batch Analysis (Multiple Images)

- To check multiple images at once, specify the folder path containing the images:

### Command Structure:

```
java -jar StegExpose.jar <folder_path>
```

## 7. Advanced Options (Optional)

- To view additional parameters, such as options for adjusting detection sensitivity or output verbosity, use the --help flag:

### Command:

```
java -jar StegExpose.jar --help
```

## 8. Review the Results

- Review the scores generated for each image and use the threshold values to determine which images require further forensic investigation to extract the suspected hidden data.

### Result:

The hidden data within the image was successfully detected using StegExpose. The analysis revealed that images with a suspect score above the threshold likely contain embedded steganographic content, confirming the tool's effectiveness in steganography detection.